

ER diagram to Relational schema

```
CREATE TABLE TimePeriod (  
    FL_DATE DATE PRIMARY KEY,  
    YEAR SMALLINT NOT NULL,  
    MONTH VARCHAR(20) NOT NULL,  
    DAY_OF_MONTH TINYINT NOT NULL,  
    DAY_OF_WEEK VARCHAR(20) NOT NULL,  
    QUARTER VARCHAR(20) NOT NULL  
);
```

```
CREATE TABLE Airlines(  
    OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,  
    OP_UNIQUE_CARRIER VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE Airports(  
    AIRPORT_ID INT PRIMARY KEY,  
    IATA_CODE VARCHAR(20) NOT NULL,  
    CITY_NAME VARCHAR(80) NOT NULL,  
    STATE_ABR VARCHAR(20) NOT NULL,  
    FOREIGN KEY (STATE_ABR) REFERENCES State(STATE_ABR)  
);
```

```
CREATE TABLE Flights (  
    FLIGHT_ID INT PRIMARY KEY,  
    OP_CARRIER_FL_NUM INT NOT NULL,  
    TAIL_NUM VARCHAR(20) NULL,  
    OP_CARRIER_AIRLINE_ID INT NOT NULL,  
    ROUTE_ID INT NOT NULL,  
    FL_DATE DATE NOT NULL,  
    CANCELLED TINYINT NOT NULL,  
    DIVERTED TINYINT NOT NULL,  
    FOREIGN KEY (OP_CARRIER_AIRLINE_ID) REFERENCES  
    Airlines(OP_CARRIER_AIRLINE_ID),  
    FOREIGN KEY (ROUTE_ID) REFERENCES Routes(ROUTE_ID), FOREIGN KEY  
(FL_DATE) REFERENCES TimePeriod(FL_DATE)  
);
```

```
CREATE TABLE FlightPerformance (  
    FLIGHT_ID INT PRIMARY KEY,  
    CRS_DEP_TIME TIME NOT NULL,  
    DEP_TIME TIME,
```

```

    DEP_DELAY INT,
    DEP_DELAY_NEW INT,
    DEP_DEL15 TINYINT,
    TAXI_OUT SMALLINT,
    WHEELS_OFF TIME,
    WHEELS_ON TIME,
    TAXI_IN SMALLINT,
    CRS_ARR_TIME TIME,
    ARR_TIME TIME,
    ARR_DELAY INT,
    ARR_DELAY_NEW INT,
    ARR_DEL15 TINYINT,
    CRS_ELAPSED_TIME SMALLINT,
    ACTUAL_ELAPSED_TIME SMALLINT,
    AIR_TIME SMALLINT,
    CANCELLATION_CODE CHAR(50),
    DIV_AIRPORT_LANDINGS TINYINT,
    DIV_REACHED_DEST TINYINT,
    DIV_ACTUAL_ELAPSED_TIME TINYINT,
    DIV_ARR_DELAY INT,
    DIV_DISTANCE INT,
    FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID),
    FOREIGN KEY (CANCELLATION_CODE) REFERENCES
    CancellationReason(CANCELLATION_CODE)
);

CREATE TABLE Routes (
    ROUTE_ID INT PRIMARY KEY AUTO_INCREMENT,
    ORIGIN_AIRPORT_ID INT NOT NULL,
    DEST_AIRPORT_ID INT NOT NULL,
    DISTANCE INT NOT NULL,
    FOREIGN KEY (ORIGIN_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID), FOREIGN
    KEY (DEST_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID)
);

CREATE TABLE FlightDelay (
    DELAY_ID TINYINT NOT NULL,
    FLIGHT_ID INT NOT NULL,
    MINUTES INT,
    PRIMARY KEY (DELAY_ID, FLIGHT_ID),
    FOREIGN KEY (DELAY_ID) REFERENCES DelayType(DELAY_ID), FOREIGN KEY
    (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID)
);

```

);

```
CREATE TABLE FlightDiversions (
    FLIGHT_ID INT NOT NULL,
    DIVERSION_NUMBER TINYINT NOT NULL,
    DIV_AIRPORT_ID INT NOT NULL,
    DIV_TOTAL_GTIME SMALLINT NULL,
    PRIMARY KEY (FLIGHT_ID, DIVERSION_NUMBER),
    FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID), FOREIGN KEY
(DIV_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID)
);
```

```
CREATE TABLE StateNames (
    STATE_ABR CHAR(2) PRIMARY KEY,
    STATE_NM VARCHAR(50) NOT NULL
);
```

```
CREATE TABLE AirlineNames (
    OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,
    NAME VARCHAR(150) NOT NULL
);
```

```
CREATE TABLE AirportNames (
    AIRPORT_ID INT PRIMARY KEY,
    NAME VARCHAR(100) NOT NULL
);
```

```
CREATE TABLE DelayType (
    DELAY_ID TINYINT PRIMARY KEY,
    NAME VARCHAR(50) NOT NULL
);
```

```
CREATE TABLE CancellationReason (
    CANCELLATION_CODE CHAR(1) PRIMARY KEY, DESCRIPTION VARCHAR(50) NOT
NULL
);
```